How are they using IIoT?

- ***Panasonic.*** Panasonic India has implemented IIoT solutions for manufacturing consumer electronics under Panasonic and Anchor brand names. It is also in the process of deploying the same for the automotive solutions it offers.
- ***Deki.*** Deki Electronics has tried implementing industry 4.0 for the cosmetic checks of capacitors it produces. The team at Deki Electronics now comprises Industry 4.0 experts, who in collaboration with the manufacturing team, are busy decoding the areas where they can implement IIoT and how they can do it when it comes to legacy machines.
- ***Lava.*** Lava International is in the process of implementing industry 4.0 at all stages of manufacturing and testing. The company's focus is to enable humans to do more innovative work and hand over the repetitive processes to machines.
- ***AT&S.*** AT&S, at its facility in India, has deployed IIoT/Industry 4.0 for predictive maintenance. It has also been able to successfully deploy the same for traceability. The company has plans to deploy IIoT at a larger scale within the next five years.

offer outlays on how to implement IIoT for small companies like us, the cost factor often scares us. Till now, we have had limited access to IIoT. We have also learned to stay away from implementing IIoT in some areas. We tried to do visual inspections of millions of capacitors we manufacture every day, and we learned that the technology has not matured enough to be implemented there," he adds.

Sharma also notes that the current pandemic is accelerating the adoption of IIoT as the economies have started looking beyond just one source for their manufacturing needs. In his regard, the IIoT can further accelerate the process of making more manufacturing hubs in the world, as it can bring manufacturing near places where the demand is on the higher side.

"It might not mean that things will remain positive for SMEs. But we need to carefully place our bets on the right parts of IIoT. We need to embrace the change for good," he says.

Let's get started

To start with, there exist machines that are capable of working 24 hours a day and seven days a week, without taking any breaks. Next is the reduced error in manufacturing owing to the data collected and fed in such machines. Then comes long-term data and its analysis. The same can help in managing better productivity shifts and better quality products.

A report by IDC had highlighted that an increase in the quality of products manufactured helps in lowering down the amount of waste generated. It is a simple fact that lower waste generated is directly proportional to lower costs for manufacturing.

"Whether you implement IIoT fully or at some stages, the same promises efficiency, accuracy, and capability. Now some people can claim that they have implemented IIoT fully, but there is still a long way to go. I see these three things as the biggest advantages of implementing IIoT," notes Sanjeev Agarwal, chief manufacturing officer, Lava International.

He adds, "The three factors have been important for every manufacturer since the inception of manufacturing. However, with increased competition and increased demand from customers, these have become critical now. IIoT can help in reducing costs, making products better, and increasing profits."

An MPI (composed of the Management Performance Institute, the Marketing Performance Institute, and the Manufacturing Performance Institute) Internet of Things Study (2017), sponsored by Binder Dijker Otte (BDO), found that manufacturers in the United States are making headway toward embracing IoT and improving their readiness—more than half (51 per cent) characterise themselves as IoT-competitive companies, and another fourteen per cent say that they're IoT leaders. Seventy two per cent of manufacturers (who contributed to the study) reported that the application of IoT in their plants and processes resulted in increased productivity last year, while sixty nine per cent reported an increase in profitability, with twelve per cent reporting an increase of more than ten per cent.

On where to start with implementing IIoT, Sanjeev Agarwal says, "Processes of manufacturing requiring repetitive tasks should be handed over to machines. Humans involved in manufacturing processes should be more involved in creating value. They should be working on innovating new processes or making the existing processes better."

Challenges in adopting Industry 4.0

Despite the proven advantages, the adoption of IIoT has not been that promising, especially in India. Not just India, a recent report by IDC noted that the worldwide spending on the Internet of Things (IoT) has been significantly impacted by the economic effects of the Covid-19 pandemic in 2020, although a back to double-digit growth rebound is expected both in the mid and long-term.

A new update to the International Data Corporation (IDC) Worldwide Internet of Things Spending Guide shows IoT spending growing 8.2 per cent year over year to 742 billion dollars in 2020, down from 14.9 per cent growth forecast in November 2019 release.

Nevertheless, IDC expects that global IoT spending will return to double-digit growth rates in 2021 and achieve a CAGR of 11.3 per cent over the 2020-2024 forecast period.

"The very first challenge in adopting IIoT is the lack of proper standardisation. The same requires a lot of work and heavy lifting. Secondly, if you look at it from a machine-to-machine point of view, it becomes difficult to get a complete set of data from the two. The challenge amplifies when you look at the legacy machines' point of view," notes Manish Misra.

The number of SMEs and big companies using legacy machines for manufacturing is huge. Embedding such machines with sensors or data loggers is as big a challenge as it is an opportunity. One of the main reasons that manufacturers do not switch to new machines is the feeling that legacy machines are easy to repair. Additionally, the cost of replacing a whole system of a legacy machine with new ones sounds a little on the expensive side to many.